NLP Fundamentals Interview Revision Cheatsheet

This cheatsheet provides a high-density, 1-page reference of core formulas, computational complexities, variable legends, and troubleshooting configurations for technical interviews.

1. Component Variable Denotation Legend

Symbol	Definition	Symbol	Definition
N	Total number of documents in the corpus	d	Model hidden / embedding dimension size
L	Token sequence length (context window)	K	Number of negative samples in SGNS
$\|V\|$	Vocabulary size (number of unique tokens)	$C(x)$	Count/frequency of token x
c	Candidate token sequence length (metrics)	r	Reference token sequence length (metrics)
k_1	BM25 term frequency saturation scaling parameter	b	BM25 document length penalty parameter

2. Computational Complexity Matrix

Algorithm	Training Time	Inference Time	Space (Memory)
Bag-of-Words	$O(N \cdot L)$	$O(L)$	$O(N \cdot \|V\|)$ sparse matrix
TF-IDF Vectorization	$O(N \cdot L + \|V\|)$	$O(L)$	$O(N \cdot \|V\|)$ sparse matrix
N-gram Language Model	$O(N \cdot L)$	$O(1)$ lookup	$O(\|V\|^N)$ transition table
Recurrent Neural Net (RNN)	$O(L \cdot d^2)$	$O(L \cdot d^2)$ (Sequential)	$O(d^2)$ parameter weights
Long Short-Term Memory (LSTM)	$O(L \cdot d^2)$	$O(L \cdot d^2)$ (Sequential)	$O(4 \cdot d^2)$ parameter weights
Gated Recurrent Unit (GRU)	$O(L \cdot d^2)$	$O(L \cdot d^2)$ (Sequential)	$O(3 \cdot d^2)$ parameter weights
Scaled Dot-Product Attention	$O(L^2 \cdot d)$	$O(L^2 \cdot d)$ (Parallel)	$O(L^2)$ intermediate attention matrix
BLEU-N Evaluation	-	$O(c \cdot r)$	$O(c)$ tokens
BERTScore Evaluation	-	$O(c \cdot r \cdot d) + \text{Forward Pass}$	$O(c \cdot r)$ similarity table

3. Master Formula Reference

Vector Semantics & Retrieval:

- **Smooth IDF**: $\text{IDF}(t, D) = \ln\left(\frac{1+N}{1+\text{DF}(t)}\right) + 1$
- L_2 **Normalization**: $\mathbf{v}_{\text{norm}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$
- **Okapi BM25 Score**: $\text{Score}(d, q) = \sum_{t \in q} \ln\left(\frac{N - \text{DF}(t) + 0.5}{\text{DF}(t) + 0.5} + 1\right) \times \frac{\text{TF}(t,d) \cdot (k_1 + 1)}{\text{TF}(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}$

Statistical & Smoothing Models:

- **Laplace (Add-One) Bigram**: $P_{\text{Laplace}}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}, w_i) + 1}{C(w_{i-1}) + |V|}$
- **Sequence Perplexity**: $\text{PPL}(W) = P(w_1, \ldots, w_m)^{-\frac{1}{m}} = e^{\mathcal{L}_{\text{CE}}}$

Continuous Word Embeddings:

- **SGNS Loss**: $\mathcal{L}_{\text{SGNS}} = -\ln \sigma(\mathbf{v}_w \cdot \mathbf{v}'_{w_c}) - \sum_{i=1}^{K} \ln \sigma(-\mathbf{v}_w \cdot \mathbf{v}'_{w_i})$

- **Noise Sampling Distribution**: $P_n(w) \propto U(w)^{0.75}$

Sequence Models & Attention:

- **LSTM CEC derivative**: $\frac{\partial \mathbf{C}_t}{\partial \mathbf{C}_{t-1}} \approx \mathbf{f}_t \approx \mathbf{1}$ (when forget gate is open, bypassing recurrent weight vanishing decay)
- **Scaled Dot-Product Attention**: $\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$
- **BLEU Brevity Penalty**: $\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \le r \end{cases}$

Production Drift Monitoring:

- **Population Stability Index**: $\text{PSI} = \sum_{i=1}^{B} (P_i - Q_i) \times \ln\left(\frac{P_i}{Q_i}\right)$ (stable: < 0.1, moderate: $0.1\text{–}0.25$, high: > 0.25)

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4. Top 5 Critical Troubleshooting Guidelines

1. **Vanishing Gradients in Recurrent State Paths**:
   - *Problem*: Backpropagated gradients vanish due to multiplicative chain products (W_{hh}^{T-t}).
   - *Fix*: Use LSTMs/GRUs where cell state CEC updates are additive, or switch to Transformer self-attention.
2. **Softmax Saturation in Self-Attention**:
   - *Problem*: For large key sizes d_k, query-key dot products blow up, pushing Softmax inputs into flat regions where derivative is near 0.
   - *Fix*: Divide attention scores by $\sqrt{d_k}$ to scale variance to 1.0, keeping gradients sensitive.
3. **Exposure Bias in Seq2Seq Models**:
   - *Problem*: Decoders are trained using Teacher Forcing (feeding true tokens), but generate auto-regressively at inference (feeding model's own outputs), compounding errors.
   - *Fix*: Apply Scheduled Sampling (randomly feeding model predictions during training) or use sequence-level reinforcement learning objectives.
4. **Data (Covariate) Shift vs. Concept Drift**:
   - *Problem*: Model accuracy decays in production because vocabulary characteristics shift ($P(X_{\text{prod}}) \neq P(X_{\text{train}})$) or meanings shift ($P(Y \mid X_{\text{prod}}) \neq P(Y \mid X_{\text{train}})$).
   - *Fix*: Monitor Population Stability Index (PSI) or Wasserstein Distance on live logs; trigger retrain loops when PSI > 0.25.
5. **Vocabulary Serving Crash**:
   - *Problem*: Text normalization and vocabulary indexing tables mismatch between the preprocessing serving API and model container, leading to corrupted embeddings.
   - *Fix*: Package tokenizer configurations and token dictionaries directly as read-only assets in the model registry alongside model weights.